

Panagiotis Theodoropoulos

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Research Interests

Deep generative models, (Entropic) Optimal Transport, Dynamic Transport maps, Dynamical Systems, Differential Equations, Schrodinger Bridge, Stochastic Optimal Control, Robust Optimal Control, Scalable higher-order optimization, Dynamic Programming, Hamilton-Jacobi Bellman principle.

Education

Georgia Institute of Technology, Atlanta, GA
PhD in Machine Learning 2022–
Advisor: [Dr. Evangelos Theodorou](#)

University of Patras, Patras, Greece
B.S.- M.S. in Mechanical Engineering (GPA: 9.00/10.00, Ranked 1st in Class of 2021) 2016–2021
Advisor: [Dr. Spilios Fassois](#)

Research Experience

Morgan Stanley, New York, NY
Machine Learning Research Associate (Advisor: [Wei Deng](#)) June–August 2026

Prisma Electronics, Athens, Greece
Machine Learning Engineer (Supervisor: [Christos Spandonidis](#)) 10/2020–06/2022

Prisma Electronics, Athens, Greece
Machine Learning Engineer Intern (Supervisor: [Christos Spandonidis](#)) Summer of 2020

Stochastic Mechanical Systems & Automation, Patras, Greece
Undergraduate Research Associate (Supervisor: [Fassois Spilios](#)) 09/2019 - 07/2021

Publications

(*Equal Contribution, [†]Equal Advising)

Preprints

- [P1] **Theodoropoulos, P.**, Jiang, N., Duan, X., Hasan, A., Nevmyvaka, Y., Theodorou, E. A., & Deng, W.
Explore Broadly, Reason Sharply: Push Small Models toward the Frontier via Sampling.
arXiv preprint arXiv:2609.38104 (2026).

Conferences

- [C1] **Theodoropoulos, P.**, Nam, J., Theodorou, E. A., & Choi, J.
Generalized Schrödinger Bridge on Graphs.
International Conference on Machine Learning (ICML) 2026.

- [C2] **Theodoropoulos, P.**, Saravanos, A. D., Theodorou, E. A.[†], & Liu, G.-H.[†]
Momentum Multi-Marginal Schrödinger Bridge Matching.
Neural Information Processing Systems (NeurIPS) 2025
- [C3] **Theodoropoulos, P.**, Komianos, N., Pacelli, V., Liu, G.-H., & Theodorou, E. A.
Feedback Schrödinger Bridge Matching. [Oral, top 1.0%]
International Conference on Learning Representations (ICLR) 2025.
- [C4] **Theodoropoulos, P.**, Liu, G.-H., Chen, T., Saravanos, A. D., & Theodorou, E. A.
A Robust Differential Neural ODE Optimizer.
International Conference on Learning Representations (ICLR) 2024.
- [C5] **Theodoropoulos, P.**, Spandonidis, C., Giordamlis, C. & Fassois, S.
Stability and Safety of Ships and Ocean Vehicles: Enhancing vessel's operational safety using Convolutional Neural Networks and Big Data techniques.
International Conference on the Stability and Safety of Ships and Ocean Vehicles (STAB&S) (2021)
- [C6] **Theodoropoulos, P.**, Spandonidis, C., Giordamlis, C.
Combined multi-layered big data and responsible AI techniques for enhanced decision support in Shipping.
International Conference on Decision Aid Sciences and Application (DASA) (2020)

Journals

- [J1] **Theodoropoulos, P.**, Spandonidis, C. C., & Fassois, S.
Use of Convolutional Neural Networks for vessel performance optimization and safety enhancement. *Ocean Engineering*, (2022).
- [J2] Spandonidis, C. C.[★], **Theodoropoulos, P.[★]**, Giannopoulos, F., Galiatsatos, N. & Petsa, A.
Evaluation of deep learning approaches for oil & gas pipeline leak detection using wireless sensor networks. *Engineering Applications of Artificial Intelligence*, (2022).
- [J3] **Theodoropoulos, P.**, Spandonidis, C. C., Giannopoulos, F. & Fassois, S.
A deep learning-based fault detection model for optimization of shipping operations and enhancement of maritime safety. *Sensors* (2022).
- [J4] **Theodoropoulos, P.**, Spandonidis, C. C., Themelis, N., Giordamlis, C. & Fassois, S.
Evaluation of Different Deep-Learning Models for the Prediction of a Ship's Propulsion Power.
Journal of Marine Science and Engineering, (2021).
- [J5] Spandonidis, C., Giannopoulos, F., Sedikos, E., Reppas, D. & **Theodoropoulos, P.**
Development of a MEMS-based IoV system for augmenting road traffic survey.
IEEE Transactions on Instrumentation and Measurement, (2022).

Workshops

- [W1] **Theodoropoulos, P.**, Liu, G.-H., Chen, T. & Theodorou, E.
Game Theoretic Neural ODE Optimizer.
In ICML Workshop on New Frontiers in Learning, Control, and Dynamical Systems (2023).

Invited Talks

- **Morgan Stanley ML Research Speaker Series** — *Stochastic Optimal Control and Optimal Transport for Generative Modeling.*

Honors & Awards

- Awarded by 'Techniko Epimelitirio Elladas - TEE' for graduating first in undergraduate studies
- Gerontelis Foundation Doctoral Scholar
- Salutatorian, Graduating Class of 2021, Department of Mechanical Engineering, University of Patras
- Eurobank scholar during undergraduate studies
- Eleytherios Venizelos scholar during undergraduate studies

Peer Review Service

Conferences:

- International Conference on Machine Learning (ICML), since 2025
- International Conference on Learning Representations (ICLR), since 2025
- AAAI Conference on Artificial Intelligence (AAAI), since 2026

Journals:

- Transactions on Machine Learning Research (TMLR), since 2026
- Engineering Applications of Artificial Intelligence
- Ocean Engineering
- Sensors
- International Journal of Maritime Engineering

Teaching Experience

Head Teaching Assistant, Georgia Institute of Technology

Spring 2025

COE 2001: Statics

- Led weekly recitations for 50+ students
- Assisted in course coordination, homework grading rubric preparation, and grading.

Skills

Programming: Python, Matlab, C/C++
Packages: Pytorch, Tensorflow, CVXPY